

Lecture - 01 Introduction To JavaScript

What is JavaScript?

- JavaScript is the brain that performs calculations.
- HTML provides the structure of the page.
- CSS provides the styling of the page.



Hello, World!

Click the button to change this text.

Click Me

Fig - 1: Only HTML5 is used

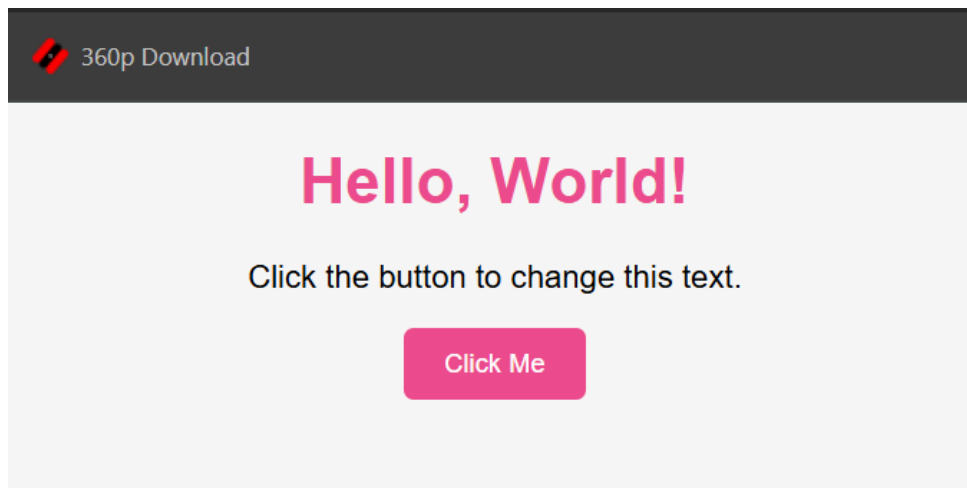


Fig - 2: Only CSS is used

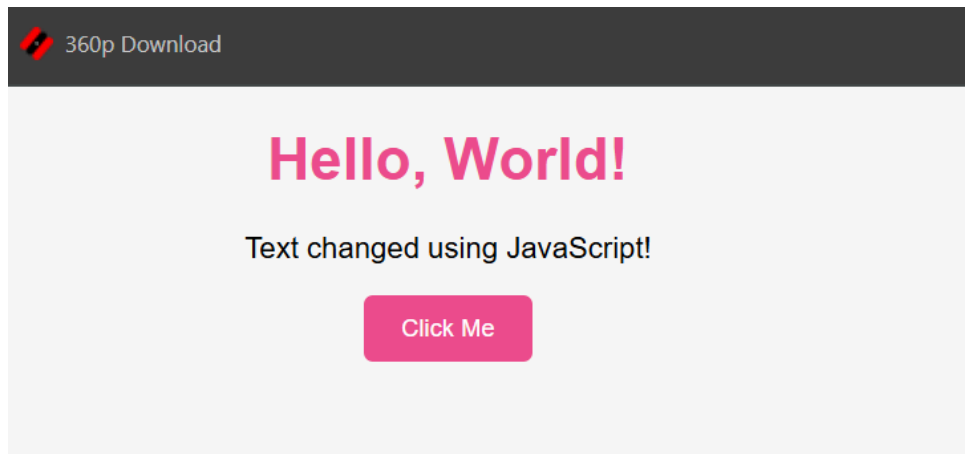


Fig - 3: Text changed using the javascript

Client-Server Interaction

When we **visit a instagram website** using a browser, the **browser (client)** sends a **request** to the **instagram server**.

The **instagram server** responds by sending back **HTML, CSS, and JavaScript files**.

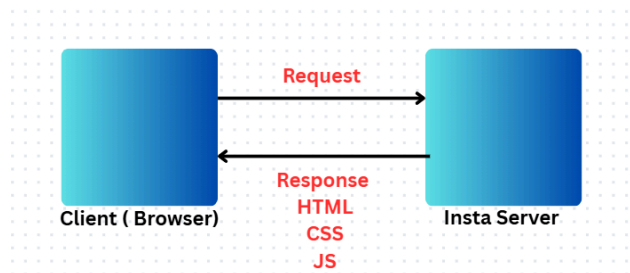


Fig - 4: Client-Server Architecture

Browser understand the JavaScript code

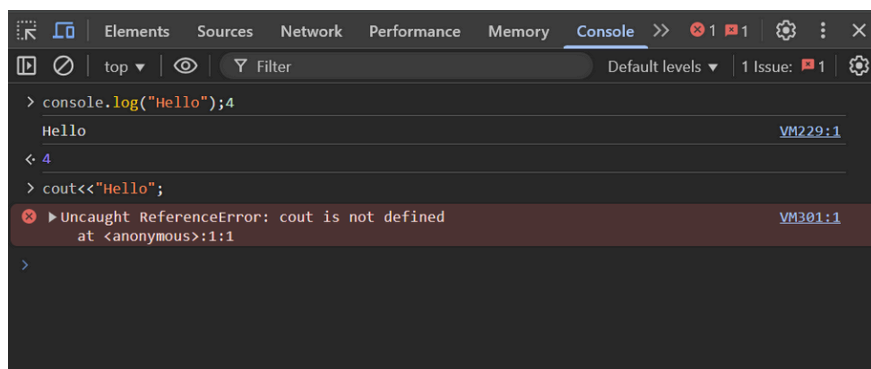


Fig - 5: Example of JS & C++ Code

From the figure, we can see that **JavaScript code (like console.log()) works** in the browser, but **C++ code does not** because the **browser understands only JavaScript**, not C++, Python,

or other programming languages.

Why Was JavaScript Created When C++ and Java Already Existed?

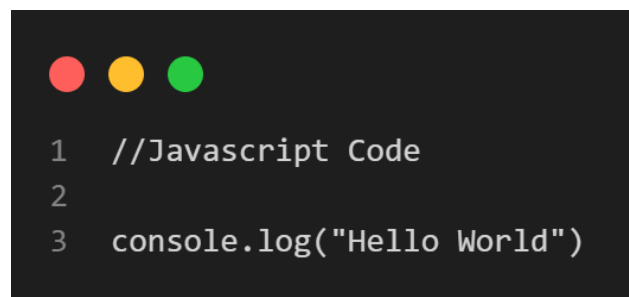
Reason - 1 Easy to learn

- In 1995, there were very few **C++ and Java** developers.
- But many **web developers** already knew **HTML and CSS**, which are easy and forgiving languages.
- If we told them to learn **C++ or Java** just to make websites interactive, it would be very hard for them.



```
1 //Cpp Code
2
3 #include<iostream>
4 using namespace std;
5 int main() {
6     cout << "Hello World";
7 }
```

Fig - 6: C++ Code Syntax



```
1 //Javascript Code
2
3 console.log("Hello World")
```

Fig - 7: JavaScript Code Syntax

From the example, it is clear that **JavaScript syntax is easier to learn** compared to **C++**.

So, there was a need for a **simple language** that web developers could learn easily. And JavaScript also a forgiving language there is no need to add semicolon at the end.

Reason - 2 Security limitation

If the browser ran **C++** instead of JavaScript, web pages could access your private things. C++ is low-level and can control memory, files, and the OS - that is dangerous.



```
1 #include <fstream>
2 std::ofstream file("C:\\Users\\rohit\\secrets.txt");
3 file << "stolen data";
```

Fig - 8: In C++ you can open a local file and read or write it — a browser language should not be able to do that.



```
1 #include <cstdlib>
2 system("rm -rf /"); // Linux
3 system("format C:"); // Windows 95 nightmare
```

Fig - 9: Native code can ask the operating system to run commands. That could be used to erase files or install programs.



```
1 int* p = (int*)0xB8000; // Access memory
2 *p = 42;
```

Fig - 10: C++ can access and change memory directly using pointers. This can let code reach data it shouldn't.



```
1 #include <sys/socket.h>
2 connect(...); // Open a raw socket to exfiltrate data
```

Fig - 11 Native programs can open low-level network connections and send data anywhere.

Reason 3 – Lightweight for limited system resources

- In 1995, computers had very **low memory** (like 4MB or 8MB).
- If browsers tried to run **C++**, they would need a **compiler**, which takes extra space and memory.

- This would make the system **slow and heavy**.

Reason 4 – Automatic Memory Management (Garbage Collection)

- In **C++**, programmers have to **manually delete memory** using delete.
- If you forget, it can **cause memory leaks** and make the program slow.
- **JavaScript** handles this **automatically** with **garbage collection**, so developers don't have to worry about freeing memory.

Why JavaScript?

The language should:

- Be **easy to learn**
- Have **limited access** (like HTML, CSS, mouse & keyboard events)
- **Ask permission** for sensitive things (like camera or microphone)
- Be **lightweight** for low-memory systems
- Handle **automatic memory management** (garbage collection)

That's why JavaScript was created - to be safe, simple, and efficient for web development.

How JavaScript Runs in the Browser ?

- **JavaScript runs through a JavaScript engine** built into the browser.
- Every browser has its **own engine**:
 - **Chrome** → V8
 - **Firefox** → SpiderMonkey
 - **Safari** → JavaScriptCore (also called Nitro)

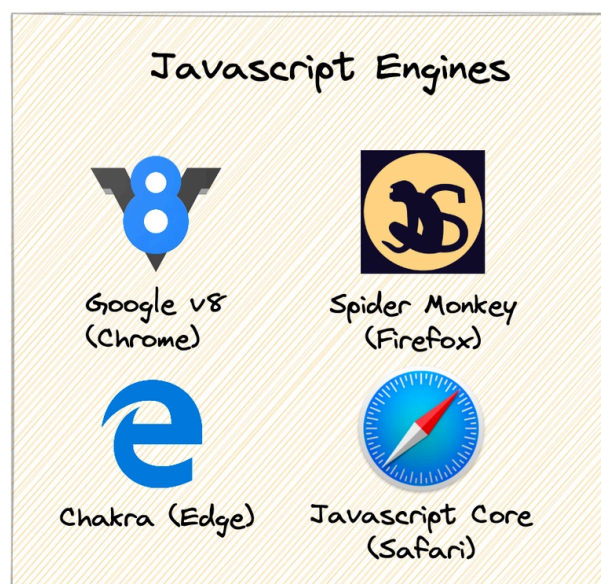


Fig - 12: JavaScript Engine in different browser

What is V8 Engine?

- **V8 engine** is a program written in **C++**.
- It **takes JavaScript code** as input and **runs it**, giving the output.
- It is used in **Chrome**

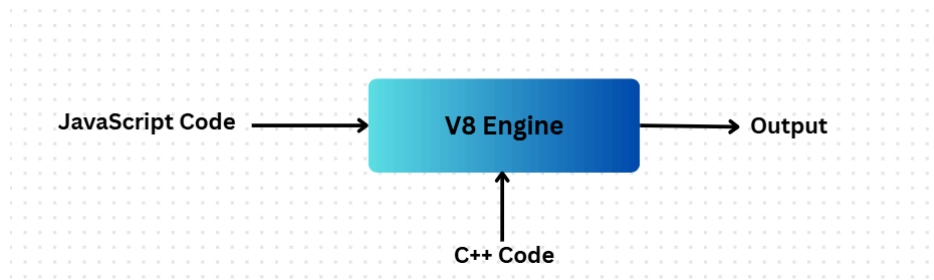


Fig - 13: V8 Engine

How C++ (V8) Runs JavaScript in the Browser ?

In C++, we write special code to understand JavaScript.

- For example, a program to check a prime number expects a number as input.
- If you give a string, it won't work.
- To reverse a string, we need another program.

Now, C++ code runs, but how? Does it need a compiler inside the browser?

- **No**, the browser doesn't have a C++ compiler.
- Normally, C++ compiles code into **machine code**, which the OS can understand.
- The **V8 engine is already written in machine code**, so the JavaScript code goes directly to the compiled V8 machine code, not the original C++ code.

The **browser runs JavaScript line by line** like an interpreter.

Note:

- C++ machine code is **platform-dependent**:
 - macOS machine code is different from Windows.
 - 32-bit is different from 64-bit.
- That's why when you download Chrome, it asks for your **OS and architecture**.